Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

IT13: ADBMS

Chapter 2:SQL

Practice Assignment Solution Template

Write a query and paste a screenshot of your output below the question.

1. Create a database named CompanyDB.
2. Create a table Employees with the following columns:
   Employees (EmpID (Integer, Primary Key), EmpName, Salary, JoinDate)
3. Create a table Departments with:

   Departments (DeptID (Integer, Primary Key), DeptName)
4. Add a column Email (VARCHAR(100)) to the Employees table.
5. Add a column dept_location to the Departments table.
6. Add a NOT NULL Constraint to the Phone column
7. Change the size of the EmpName column from VARCHAR (50) to VARCHAR (100).
8. Drop the primary key Constraint from Departments
9. Create a foreign key constraint between two tables
10. Rename the column EmpName to EmployeeName.
11. Apply a CHECK constraint on the Age and Salary columns.
12. Remove the Phone column from the Employees table.
13. Rename the table Employees to EmployeeDetails.
14. Drop the CHECK constraints.
15. Delete the Departments table permanently.
16. Drop the database.

1. **create database CompanyDB1;**

   show databases;

```
mysql>
mysql> create database CompanyDB;
ERROR 1007 (HY000): Can't create database 'companydb'; database exists
mysql> create database CompanyDB1;
Query OK, 1 row affected (0.17 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| adbms_practice     |
| companydb          |
| companydb1         |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
7 rows in set (0.05 sec)

mysql>
```

2. **USE CompanyDB1;**
   Database changed
   mysql> CREATE TABLE Employees1(
   -> EmpID INT PRIMARY KEY,
   -> EmpName VARCHAR(50),
   -> Salary DECIMAL(10,2),
   -> JoinDate DATE
   ->);

   DESCRIBE Employees1;

```
mysql> DESCRIBE Employees1;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EmpID    | int          | NO   | PRI | NULL    |       |
| EmpName  | varchar(50)  | YES  |     | NULL    |       |
| Salary   | decimal(10,2)| YES  |     | NULL    |       |
| JoinDate | date         | YES  |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
```

3.

USE CompanyDB1;
Database changed
mysql> create table Departments1(
-> DeptID int primary key,
-> DeptName varchar(100)
->);

DESCRIBE Departments1;

```
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| DeptID   | int          | NO   | PRI | NULL    |       |
| DeptName | varchar(100) | YES  |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
```

4.

USE CompanyDB1;
Database changed
mysql> alter table Employees1
-> add Email varchar(100);

describe Employees1;

```
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EmpID    | int          | NO   | PRI | NULL    |       |
| EmpName  | varchar(50)  | YES  |     | NULL    |       |
| Salary   | decimal(10,2)| YES  |     | NULL    |       |
| JoinDate | date         | YES  |     | NULL    |       |
| Email    | varchar(100) | YES  |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
```

5.

USE CompanyDB1;
Database changed
mysql> alter table Departments1
-> add dept_location varchar(100);

DESCRIBE Departments1;

Field	Type	Null	Key	Default	Extra
DeptID	int	NO	PRI	NULL	
DeptName	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	

6.

alter table employees1

-> add Phone varchar(15);

alter table Employees1

-> modify Phone varchar(15) not null;

describe Employees1;

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(50)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
Phone	varchar(15)	NO		NULL	

7.

```
alter table Employees1
   -> modify EmpName varchar(100);
```

describe Employees1;

```
+----------+---------------+------+-----+---------+-------+
| Field    | Type          | Null | Key | Default | Extra |
+----------+---------------+------+-----+---------+-------+
| EmpID    | int           | NO   | PRI | NULL    |       |
| EmpName  | varchar(100)  | YES  |     | NULL    |       |
| Salary   | decimal(10,2) | YES  |     | NULL    |       |
| JoinDate | date          | YES  |     | NULL    |       |
| Email    | varchar(100)  | YES  |     | NULL    |       |
| Phone    | varchar(15)   | NO   |     | NULL    |       |
+----------+---------------+------+-----+---------+-------+
```

8.

```
alter table Departments1
   -> drop primary key;
```

DESCRIBE Departments1;

```
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| DeptID        | int          | NO   |     | NULL    |       |
| DeptName      | varchar(100) | YES  |     | NULL    |       |
| dept_location | varchar(100) | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
```

9.

```
mysql> USE CompanyDB1;
Database changed
mysql> alter table Employees1
   -> add DeptID int;
```

mysql> alter table Departments1
-> add primary key (DeptID);

mysql> alter table Employees1
-> add constraint FK_emp_dep
-> foreign key (DeptID)
-> references Departments1(DeptID);

mysql> describe Employees1;

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(100)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
Phone	varchar(15)	NO		NULL	
DeptID	int	YES	MUL	NULL	

mysql> show create table Employees1;

```
| Employees1 | CREATE TABLE `employees1` (
  `EmpID` int NOT NULL,
  `EmpName` varchar(100) DEFAULT NULL,
  `Salary` decimal(10,2) DEFAULT NULL,
  `JoinDate` date DEFAULT NULL,
  `Email` varchar(100) DEFAULT NULL,
  `Phone` varchar(15) NOT NULL,
  `DeptID` int DEFAULT NULL,
  PRIMARY KEY (`EmpID`),
  KEY `FK_emp_dep` (`DeptID`),
  CONSTRAINT `FK_emp_dep` FOREIGN KEY (`DeptID`) REFERENCES `departments1` (`DeptID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

10.

```
mysql> alter table Employees1
    -> rename column EmpName to EmployeeName;
```

```
mysql> describe Employees1;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(100)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
Phone	varchar(15)	NO		NULL	
DeptID	int	YES	MUL	NULL	

11.

```
mysql> alter table Employees1
    -> add Age int;
```

```
mysql> alter table Employees1
    -> add constraint ck_employee_age
    -> check (Age >=18);
```

```
mysql> alter table Employees1
    -> add constraint ck_employee_salary
    -> check (Salary >0);
```

mysql> describe Employees1;

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(100)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
JoinDate	date	YES		NULL	
Email	varchar(100)	YES		NULL	
Phone	varchar(15)	NO		NULL	
DeptID	int	YES	MUL	NULL	
Age	int	YES		NULL	

mysql> show create table Employees1;

```
| Employees1 | CREATE TABLE `employees1` (
  `EmpID` int NOT NULL,
  `EmployeeName` varchar(100) DEFAULT NULL,
  `Salary` decimal(10,2) DEFAULT NULL,
  `JoinDate` date DEFAULT NULL,
  `Email` varchar(100) DEFAULT NULL,
  `Phone` varchar(15) NOT NULL,
  `DeptID` int DEFAULT NULL,
  `Age` int DEFAULT NULL,
  PRIMARY KEY (`EmpID`),
  KEY `FK_emp_dep` (`DeptID`),
  CONSTRAINT `FK_emp_dep` FOREIGN KEY (`DeptID`) REFERENCES `departments1` (`DeptID`),
  CONSTRAINT `ck_employee_age` CHECK ((`Age` >= 18)),
  CONSTRAINT `ck_employee_salary` CHECK ((`Salary` > 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

12.

mysql> alter table Employees1

-> drop column Phone;

mysql> describe Employees1;

```
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| EmpID        | int           | NO   | PRI | NULL    |       |
| EmployeeName | varchar(100)  | YES  |     | NULL    |       |
| Salary       | decimal(10,2) | YES  |     | NULL    |       |
| JoinDate     | date          | YES  |     | NULL    |       |
| Email        | varchar(100)  | YES  |     | NULL    |       |
| DeptID       | int           | YES  | MUL | NULL    |       |
| Age          | int           | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
```

13.

mysql> rename table Employees1 to EmployeeDetails;

mysql> show tables;

```
+----------------------+
| Tables_in_companydb1 |
+----------------------+
| departments1         |
| employeedetails      |
+----------------------+
```

14.

mysql> alter table EmployeeDetails
-> drop check ck_employee_age;

mysql> alter table EmployeeDetails
-> drop check ck_employee_salary;

mysql> show create table EmployeeDetails;

```
| EmployeeDetails | CREATE TABLE `employeedetails` (
  `EmpID` int NOT NULL,
  `EmployeeName` varchar(100) DEFAULT NULL,
  `Salary` decimal(10,2) DEFAULT NULL,
  `JoinDate` date DEFAULT NULL,
  `Email` varchar(100) DEFAULT NULL,
  `DeptID` int DEFAULT NULL,
  `Age` int DEFAULT NULL,
  PRIMARY KEY (`EmpID`),
  KEY `FK_emp_dep` (`DeptID`),
  CONSTRAINT `FK_emp_dep` FOREIGN KEY (`DeptID`) REFERENCES `departments1` (`DeptID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

15.

mysql> alter table EmployeeDetails

-> drop foreign key FK_emp_dep;

mysql> drop table Departments1;

mysql> show tables;

```
+----------------------+
| Tables_in_companydb1 |
+----------------------+
| employeedetails      |
+----------------------+
```

16.

mysql> drop database CompanyDB1;

mysql> show databases;

```
+--------------------+
| Database           |
+--------------------+
| adbms_practice     |
| companydb          |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
```